PSU-48V Final Acceptance Test Procedure

ATP-0117 · revision C

⚠ **One or more files did not match their recorded checksum. See Configuration.**

UUT Type	PSU-48V
Serial	**SN240117-0042**
Project	Contract 7712 / Lot 3
Station	BENCH-03
Start	2026-10-17 14:03:22 +10:30
End	2026-10-17 14:41:09 +10:30
Duration	37 min 47 s
ATE version	1.0.3
Stop on Fail	No
Schedule	PSU-48V Final ATP Final acceptance of the 48 V 1.2 kW rectifier module after burn-in. psu48/final_atp.tslua
Media Release	MN-2041 · PSU-48V Test Media · 2.3.1

FAIL

End Reason: COMPLETED

Summary

PASS	FAIL	ERROR	SKIPPED	NOTRUN	FORCED_NOTRUN	NOREPORT
35	2	1	0	2	0	0

Retried steps: 1 · Operator enable overrides: 2

Failures

140	Insulation resistance, L-PE rail1.lua:42: attempt to compare nil with number	**ERROR**
250	Input current at no load outside limits	**FAIL**
320	Firmware version outside limits	**FAIL**

Operator

J. Citizen

Signature

Date

Steps

Step	Step Name / message	Expected	Actual	Result
10	Input current at no load within limits	0.050 .. 0.120 A	0.102 A	**PASS**
20	Output voltage, rail 1 within limits	47.50 .. 48.50 V	47.86 V	**PASS**
30	Ripple at full load within limits	< 120 mVpp	55.5 mVpp	**PASS**
40	Fan tachometer within limits	2800 .. 3600 rpm	3203 rpm	**PASS**
50	Firmware version within limits	2.14.0	2.14.0	**PASS**
60	Insulation resistance, L-PE within limits Input 1 (LED colour): Green	> 100 MΩ	231 MΩ	**PASS**
70	Efficiency at 50 % load within limits	>= 92.0 %	92.7 %	**PASS**
80	Over-temperature trip disabled by Operator	85 .. 95 °C		**NOTRUN**
90	Input current at no load within limits	0.050 .. 0.120 A	0.101 A	**PASS**
100	Output voltage, rail 1 within limits	47.50 .. 48.50 V	48.18 V	**PASS**
110	Ripple at full load within limits	< 120 mVpp	60.3 mVpp	**PASS**
120	Fan tachometer ↻ 3 within limits	2800 .. 3600 rpm	3092 rpm	**PASS**
#1	Fan tachometer outside limits → RETRY	2800 .. 3600 rpm	2710 rpm	**FAIL**
#2	Fan tachometer outside limits → RETRY	2800 .. 3600 rpm	2765 rpm	**FAIL**
130	Firmware version within limits	2.14.0	2.14.0	**PASS**
140	Insulation resistance, L-PE rail1.lua:42: attempt to compare nil with number	> 100 MΩ	645 MΩ	**ERROR**
150	Efficiency at 50 % load within limits	>= 92.0 %	92.2 %	**PASS**
160	Over-temperature trip within limits	85 .. 95 °C	93.2 °C	**PASS**
170	Input current at no load within limits	0.050 .. 0.120 A	0.088 A	**PASS**
180	Output voltage, rail 1 within limits	47.50 .. 48.50 V	47.80 V	**PASS**
190	Ripple at full load within limits	< 120 mVpp	49.5 mVpp	**PASS**

Step	Step Name / message	Expected	Actual	Result
200	Fan tachometer within limits	2800 .. 3600 rpm	3463 rpm	**PASS**
210	Firmware version within limits	2.14.0	2.14.0	**PASS**
220	Insulation resistance, L-PE within limits	> 100 MΩ	434 MΩ	**PASS**
230	Efficiency at 50 % load within limits Input 1 (LED colour): Green	>= 92.0 %	92.9 %	**PASS**
240	Over-temperature trip within limits	85 .. 95 °C	87.5 °C	**PASS**
250	Input current at no load outside limits	0.050 .. 0.120 A	0.062 A	**FAIL**
260	**Burn-in channel 1 (burnin.tslua)**			**6 steps: 5 PASS, 1 NOTRUN**
BI1.1	Output voltage, rail 1 within limits	47.50 .. 48.50 V	47.86 V	**PASS**
BI1.2	Ripple at full load within limits	< 120 mVpp	106.4 mVpp	**PASS**
BI1.3	Fan tachometer within limits	2800 .. 3600 rpm	2932 rpm	**PASS**
BI1.4	Firmware version within limits	2.14.0	2.14.0	**PASS**
BI1.5	Insulation resistance, L-PE within limits	> 100 MΩ	168 MΩ	**PASS**
BI1.6	Efficiency at 50 % load disabled by Operator	>= 92.0 %		**NOTRUN**
270	Over-temperature trip within limits	85 .. 95 °C	88.7 °C	**PASS**
280	Input current at no load within limits	0.050 .. 0.120 A	0.095 A	**PASS**
290	Output voltage, rail 1 within limits	47.50 .. 48.50 V	47.83 V	**PASS**
300	Ripple at full load within limits	< 120 mVpp	84.0 mVpp	**PASS**
310	Fan tachometer within limits	2800 .. 3600 rpm	3233 rpm	**PASS**
320	Firmware version outside limits	2.14.0	2.14.0	**FAIL**
330	Insulation resistance, L-PE within limits	> 100 MΩ	774 MΩ	**PASS**
340	Efficiency at 50 % load within limits	>= 92.0 %	93.9 %	**PASS**
350	Over-temperature trip within limits Input 1 (LED colour): Green	85 .. 95 °C	87.0 °C	**PASS**

Equipment

Name	Serial	Connection	Cal due	Notes
Digital multimeter	MY5402113	`VISA USB0::0x2A8D::0x1301::MY5402113::INSTR`	2027-03-01	
Electronic load	LD-88213	`TCP 10.0.4.21:5025`	2026-12-15	channel 2 only
Power analyser	PA3000-0193	`SERIAL COM4 115200 8N1`	2027-01-20	
Hipot tester				manual, reading entered by Operator

Configuration

Media Releases

Media Number	Name	Version	.atedef SHA-256
MN-2041	PSU-48V Test Media	2.3.1	`3b7e0c51d2a4f9e1b8c6a7d5e4f3029187a6b5c4d3e2f1a0b9c8d7e6f5a4b3c2`
MN-1100	Common Instrument Library	1.8.0	`a0f1e2d3c4b5a6978877665544332211ffeeddccbbaa998877665544332211 00`

Files

Path	Media	Expected SHA-256	Computed SHA-256	Stored
psu48/final_atp.tslua	MN-2041	`5d41402abc4b2a76b9719d911017c592ae2f6c1b77e5b2b7d1a0c9e8f7a6b5c4`	`5d41402abc4b2a76b9719d911017c592ae2f6c1b77e5b2b7d1a0c9e8f7a6b5c4`	No
psu48/burnin.tslua	MN-2041	`7c211433f02071597741e6ff5a8ea34789abad0f2c1d3e4b5a69788796a5b4c3`	`7c211433f02071597741e6ff5a8ea34789abad0f2c1d3e4b5a69788796a5b4c3`	No
lib/scpi.lua	MN-1100	`9f2c5e8a1b3d4f6a7c8e9d0b1a2c3e4f5a6b7c8d9e0f1a2b3c4d5e6f7a8b9c0d`	`41ab09f8e7d6c5b4a3928170f6e5d4c3b2a1908f7e6d5c4b3a29180f7e6d5c4b`	Yes
ate.util			`e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855`	No

Sub-schedule instances

Name	File	Prefix
Burn-in channel	psu48/burnin.tslua	BI1.
Burn-in channel	psu48/burnin.tslua	BI2.

Run log

Run		
14:03:22.101	INFO	Test Run started by J. Citizen on BENCH-03
14:03:22.140	WARNING	checksum mismatch: lib/scpi.lua (expected 9f2c…, computed 41ab…)
Pre-test Action: PSU-48V Final ATP		
14:03:22.300	INFO	power analyser zeroed
Step 10 · Input current at no load		
14:00:03.261	INFO	SCPI> MEAS:VOLT? CH1
14:00:03.460	INFO	reading stable after 3 samples
14:00:03.807	INFO	load set to 12.5 A
14:00:03.029	INFO	reading stable after 3 samples
14:00:03.780	INFO	SCPI> MEAS:VOLT? CH1
Step 20 · Output voltage, rail 1		
14:00:05.605	INFO	SCPI> MEAS:VOLT? CH1
14:00:05.031	WARNING	SCPI> MEAS:VOLT? CH1
14:00:05.554	INFO	SCPI> MEAS:VOLT? CH1
14:00:05.390	WARNING	DMM range 100 V
14:00:05.432	INFO	DMM range 100 V
Step 30 · Ripple at full load		
14:00:07.507	INFO	relay K4 closed
14:00:07.236	INFO	DMM range 100 V
14:00:07.470	INFO	settling 250 ms
14:00:07.426	WARNING	relay K4 closed
14:00:07.102	WARNING	load set to 12.5 A
Step 40 · Fan tachometer		
14:00:09.340	INFO	DMM range 100 V
14:00:09.512	WARNING	reading stable after 3 samples
14:00:09.932	INFO	DMM range 100 V
14:00:09.290	INFO	relay K4 closed
14:00:09.511	WARNING	relay K4 closed
Step 50 · Firmware version		
14:00:11.491	INFO	load set to 12.5 A
14:00:11.413	INFO	reading stable after 3 samples
14:00:11.375	INFO	relay K4 closed
14:00:11.794	INFO	DMM range 100 V
14:00:11.088	INFO	reading stable after 3 samples
Step 60 · Insulation resistance, L-PE		
14:00:13.860	INFO	reading stable after 3 samples
14:00:13.750	INFO	SCPI> MEAS:VOLT? CH1
14:00:13.315	INFO	DMM range 100 V
14:00:13.629	INFO	relay K4 closed
14:00:13.662	INFO	load set to 12.5 A
Step 70 · Efficiency at 50 % load		
14:00:15.204	INFO	relay K4 closed
14:00:15.561	WARNING	load set to 12.5 A
14:00:15.352	INFO	relay K4 closed
14:00:15.931	INFO	settling 250 ms
14:00:15.623	INFO	DMM range 100 V
Step 80 · Over-temperature trip		
14:00:17.840	INFO	DMM range 100 V
14:00:17.132	INFO	relay K4 closed

14:00:17.210	INFO	reading stable after 3 samples
14:00:17.492	WARNING	settling 250 ms
14:00:17.204	INFO	relay K4 closed

Step 90 · Input current at no load

14:00:19.001	INFO	relay K4 closed
14:00:19.805	INFO	relay K4 closed
14:00:19.614	INFO	SCPI> MEAS:VOLT? CH1
14:00:19.650	INFO	load set to 12.5 A
14:00:19.185	INFO	SCPI> MEAS:VOLT? CH1

Step 100 · Output voltage, rail 1

14:00:21.261	INFO	SCPI> MEAS:VOLT? CH1
14:00:21.689	INFO	SCPI> MEAS:VOLT? CH1
14:00:21.017	INFO	reading stable after 3 samples
14:00:21.773	INFO	settling 250 ms
14:00:21.112	INFO	relay K4 closed

Step 110 · Ripple at full load

14:00:23.261	INFO	relay K4 closed
14:00:23.672	WARNING	settling 250 ms
14:00:23.301	INFO	reading stable after 3 samples
14:00:23.508	INFO	reading stable after 3 samples
14:00:23.319	INFO	reading stable after 3 samples

Step 120 · Fan tachometer

14:00:25.259	INFO	DMM range 100 V
14:00:25.214	INFO	relay K4 closed
14:00:25.998	INFO	SCPI> MEAS:VOLT? CH1
14:00:25.406	INFO	load set to 12.5 A
14:00:25.982	INFO	load set to 12.5 A

Step 130 · Firmware version

14:00:27.436	INFO	relay K4 closed
14:00:27.999	INFO	DMM range 100 V
14:00:27.528	INFO	reading stable after 3 samples
14:00:27.664	INFO	SCPI> MEAS:VOLT? CH1
14:00:27.589	INFO	settling 250 ms

Step 140 · Insulation resistance, L-PE

14:00:29.755	INFO	settling 250 ms
14:00:29.217	INFO	SCPI> MEAS:VOLT? CH1
14:00:29.879	INFO	SCPI> MEAS:VOLT? CH1
14:00:29.961	INFO	settling 250 ms
14:00:29.426	INFO	relay K4 closed

Step 150 · Efficiency at 50 % load

14:00:31.038	INFO	relay K4 closed
14:00:31.985	INFO	relay K4 closed
14:00:31.847	INFO	DMM range 100 V
14:00:31.038	INFO	reading stable after 3 samples
14:00:31.101	INFO	load set to 12.5 A

Step 160 · Over-temperature trip

14:00:33.504	INFO	SCPI> MEAS:VOLT? CH1
14:00:33.399	WARNING	settling 250 ms
14:00:33.017	INFO	settling 250 ms
14:00:33.411	INFO	settling 250 ms
14:00:33.205	INFO	settling 250 ms

Step 170 · Input current at no load

14:00:35.439	INFO	load set to 12.5 A
14:00:35.098	INFO	reading stable after 3 samples
14:00:35.352	WARNING	DMM range 100 V
14:00:35.786	INFO	relay K4 closed
14:00:35.742	INFO	SCPI> MEAS:VOLT? CH1

Step 180 · Output voltage, rail 1

14:00:37.218	WARNING	settling 250 ms
14:00:37.614	INFO	relay K4 closed
14:00:37.376	INFO	settling 250 ms
14:00:37.298	INFO	load set to 12.5 A
14:00:37.618	INFO	DMM range 100 V

Step 190 · Ripple at full load

14:00:39.106	INFO	settling 250 ms
14:00:39.074	INFO	reading stable after 3 samples
14:00:39.807	INFO	load set to 12.5 A
14:00:39.349	INFO	SCPI> MEAS:VOLT? CH1
14:00:39.800	INFO	reading stable after 3 samples

Step 200 · Fan tachometer

14:00:41.083	INFO	settling 250 ms
14:00:41.302	INFO	relay K4 closed
14:00:41.117	INFO	reading stable after 3 samples
14:00:41.110	INFO	SCPI> MEAS:VOLT? CH1
14:00:41.012	INFO	relay K4 closed

Step 210 · Firmware version

14:00:43.117	INFO	SCPI> MEAS:VOLT? CH1
14:00:43.804	INFO	relay K4 closed
14:00:43.118	INFO	reading stable after 3 samples
14:00:43.247	INFO	load set to 12.5 A
14:00:43.105	INFO	reading stable after 3 samples

Step 220 · Insulation resistance, L-PE

14:00:45.931	WARNING	settling 250 ms
14:00:45.728	INFO	reading stable after 3 samples
14:00:45.212	INFO	DMM range 100 V
14:00:45.027	INFO	SCPI> MEAS:VOLT? CH1
14:00:45.947	INFO	settling 250 ms

Step 230 · Efficiency at 50 % load

14:00:47.408	INFO	SCPI> MEAS:VOLT? CH1
14:00:47.324	INFO	relay K4 closed
14:00:47.114	WARNING	settling 250 ms
14:00:47.632	INFO	relay K4 closed
14:00:47.480	INFO	DMM range 100 V

Step 240 · Over-temperature trip

14:00:49.203	INFO	load set to 12.5 A
14:00:49.839	INFO	settling 250 ms
14:00:49.771	INFO	reading stable after 3 samples
14:00:49.588	INFO	DMM range 100 V
14:00:49.232	INFO	reading stable after 3 samples

Step 250 · Input current at no load

14:00:51.811	INFO	relay K4 closed
14:00:51.310	INFO	load set to 12.5 A
14:00:51.557	INFO	relay K4 closed
14:00:51.610	INFO	SCPI> MEAS:VOLT? CH1
14:00:51.020	INFO	load set to 12.5 A

Step BI1.1 · Output voltage, rail 1

14:00:55.746	INFO	SCPI> MEAS:VOLT? CH1
14:00:55.010	INFO	settling 250 ms
14:00:55.367	INFO	reading stable after 3 samples
14:00:55.879	INFO	load set to 12.5 A
14:00:55.796	INFO	settling 250 ms

Step BI1.2 · Ripple at full load

14:00:57.794	INFO	load set to 12.5 A
14:00:57.841	ERROR	settling 250 ms
14:00:57.726	INFO	relay K4 closed
14:00:57.616	INFO	settling 250 ms
14:00:57.211	INFO	load set to 12.5 A

Step BI1.3 · Fan tachometer

14:00:59.840	INFO	relay K4 closed
14:00:59.928	INFO	relay K4 closed
14:00:59.764	INFO	DMM range 100 V
14:00:59.306	INFO	reading stable after 3 samples
14:00:59.049	INFO	DMM range 100 V

Step BI1.4 · Firmware version

14:01:01.796	INFO	SCPI> MEAS:VOLT? CH1
14:01:01.457	INFO	reading stable after 3 samples
14:01:01.554	INFO	reading stable after 3 samples
14:01:01.464	INFO	SCPI> MEAS:VOLT? CH1
14:01:01.346	INFO	load set to 12.5 A

Step BI1.5 · Insulation resistance, L-PE

14:01:03.426	INFO	relay K4 closed
14:01:03.708	INFO	settling 250 ms
14:01:03.607	INFO	load set to 12.5 A
14:01:03.848	INFO	settling 250 ms
14:01:03.410	INFO	load set to 12.5 A

Step BI1.6 · Efficiency at 50 % load

14:01:05.541	INFO	settling 250 ms
14:01:05.664	INFO	reading stable after 3 samples
14:01:05.654	WARNING	DMM range 100 V
14:01:05.320	INFO	reading stable after 3 samples
14:01:05.979	INFO	load set to 12.5 A

Step 270 · Over-temperature trip

14:01:07.745	INFO	DMM range 100 V
14:01:07.661	INFO	load set to 12.5 A
14:01:07.073	INFO	relay K4 closed
14:01:07.377	INFO	load set to 12.5 A
14:01:07.811	INFO	load set to 12.5 A

Step 280 · Input current at no load

14:01:09.380	INFO	load set to 12.5 A
14:01:09.754	INFO	reading stable after 3 samples
14:01:09.876	INFO	SCPI> MEAS:VOLT? CH1
14:01:09.983	INFO	relay K4 closed
14:01:09.180	INFO	load set to 12.5 A

Step 290 · Output voltage, rail 1

14:01:11.775	INFO	SCPI> MEAS:VOLT? CH1
14:01:11.403	INFO	DMM range 100 V
14:01:11.393	INFO	relay K4 closed

14:01:11.557	INFO	DMM range 100 V
14:01:11.536	ERROR	SCPI> MEAS:VOLT? CH1

Step 300 · Ripple at full load

14:01:13.934	INFO	SCPI> MEAS:VOLT? CH1
14:01:13.142	WARNING	relay K4 closed
14:01:13.675	INFO	DMM range 100 V
14:01:13.455	INFO	load set to 12.5 A
14:01:13.391	WARNING	reading stable after 3 samples

Step 310 · Fan tachometer

14:01:15.637	INFO	reading stable after 3 samples
14:01:15.122	WARNING	reading stable after 3 samples
14:01:15.418	INFO	SCPI> MEAS:VOLT? CH1
14:01:15.284	INFO	load set to 12.5 A
14:01:15.572	INFO	SCPI> MEAS:VOLT? CH1

Step 320 · Firmware version

14:01:17.592	INFO	SCPI> MEAS:VOLT? CH1
14:01:17.996	INFO	relay K4 closed
14:01:17.266	INFO	load set to 12.5 A
14:01:17.151	INFO	relay K4 closed
14:01:17.318	INFO	relay K4 closed

Step 330 · Insulation resistance, L-PE

14:01:19.881	INFO	load set to 12.5 A
14:01:19.502	INFO	reading stable after 3 samples
14:01:19.787	INFO	load set to 12.5 A
14:01:19.392	INFO	load set to 12.5 A
14:01:19.110	INFO	SCPI> MEAS:VOLT? CH1

Step 340 · Efficiency at 50 % load

14:01:21.988	INFO	DMM range 100 V
14:01:21.996	INFO	DMM range 100 V
14:01:21.512	INFO	settling 250 ms
14:01:21.318	INFO	reading stable after 3 samples
14:01:21.365	INFO	relay K4 closed

Step 350 · Over-temperature trip

14:01:23.358	INFO	settling 250 ms
14:01:23.347	INFO	DMM range 100 V
14:01:23.504	INFO	SCPI> MEAS:VOLT? CH1
14:01:23.386	INFO	reading stable after 3 samples
14:01:23.003	INFO	settling 250 ms